

Intro to Cloud Computing

The background of the slide features several thick, wavy, overlapping lines in shades of light blue, orange, and lime green, creating a dynamic and modern aesthetic.

Recap

- ▶ Implementation Level of Virtualization
- ▶ Containers VM (Docker) vs. Hypervisor VM
- ▶ Kubernetes
 - ❖ Container Orchestrator
 - ❖ Kubernetes Architecture

Implementation Level of Virtualization

► Level of Implementations

- ❖ Application
- ❖ Library Support
- ❖ Operating System
- ❖ Hardware Abstraction
- ❖ Instruction Set Architecture

What have we
learned so far?

Application level

JVM / .NET CLR / Panot

Library (user-level API) level

WINE/ WABI/ LxRun / Visual MainWin / vCUDA

Operating system level

Jail / Virtual Environment / Ensime's VPS / FVM
LXC, Docker

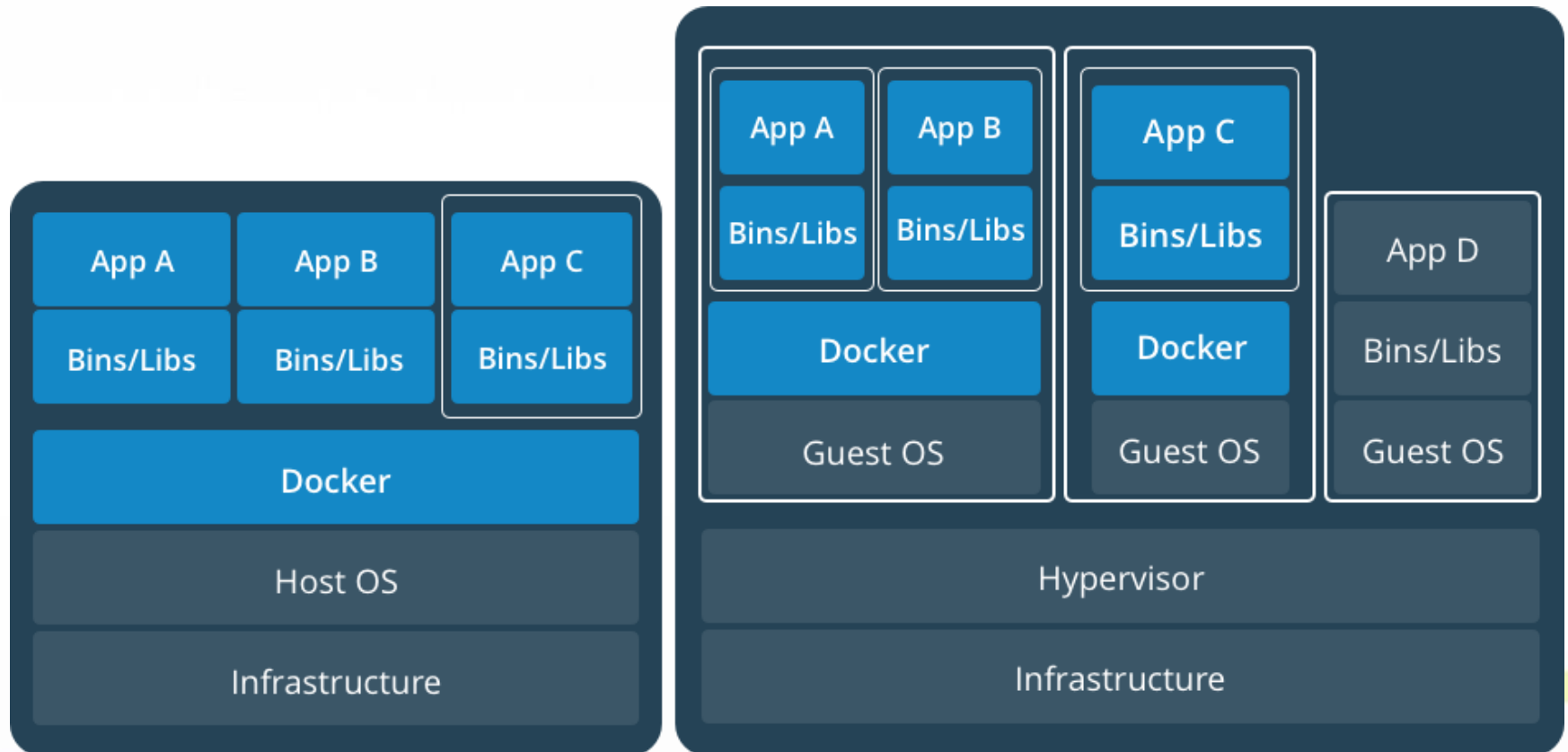
Hardware abstraction layer (HAL) level

VMware / Virtual PC / Denali / Xen / L4 /
Plex 86 / User mode Linux / Cooperative Linux

Instruction set architecture (ISA) level

Bochs / Crusoe / QEMU / BIRD / Dynamo

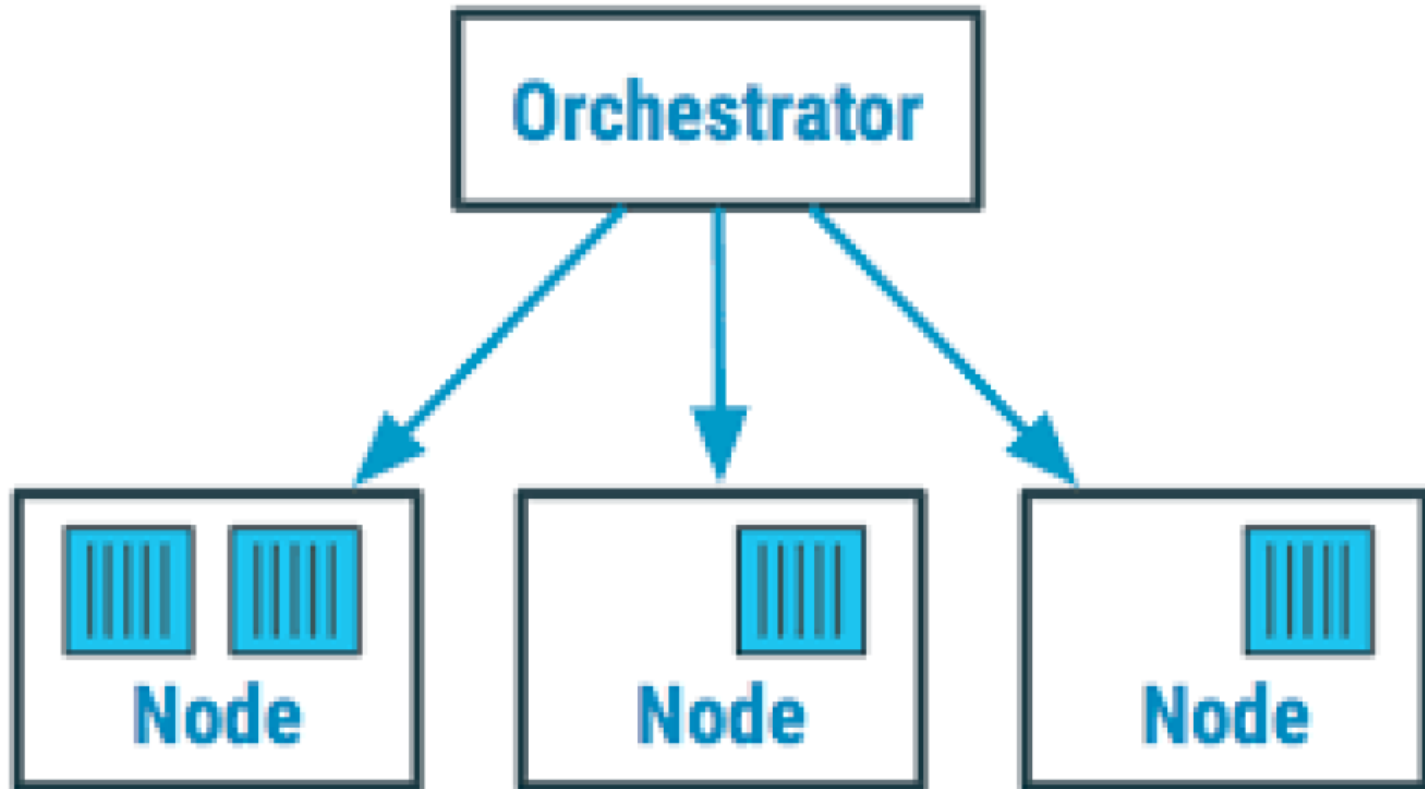
Containers VM vs. Hypervisor VM



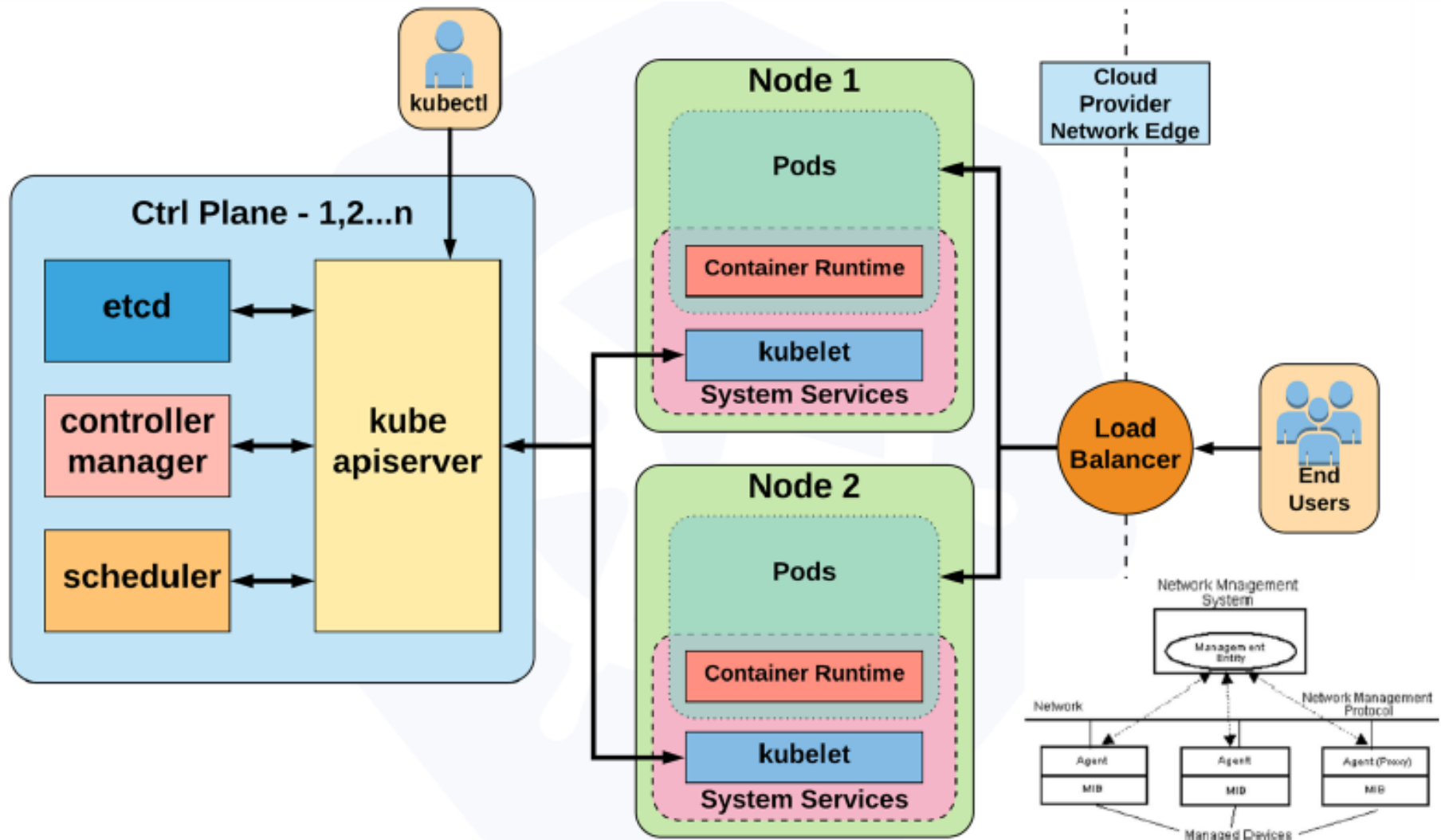
Containers (VM)

Hypervisor (VM)

Kubernetes: Container Orchestrator



Kubernetes Architecture





Contents

- ▶ Roots of Cloud Computing
 - ▶ Definition of Cloud Computing
 - ▶ Characteristics of Cloud Computing
- 

What are we trying to answer?

What is Cloud Computing?

Why Cloud Computing?

What are its challenges and opportunities?

How does Cloud Computing work?





What is Cloud Computing to User

“Computing that is hide under Cloud”



“Computing transparent to user”

System Transparencies

- ▶ Access
- ▶ Location
- ▶ Migration
- ▶ Relocation
- ▶ Replication
- ▶ Concurrency
- ▶ Failure
- ▶ Persistence



What kind of Cloud “Computing” do we need or clients want?

Existing Computing Paradigms



Basic Computing

▶ Personal Computing

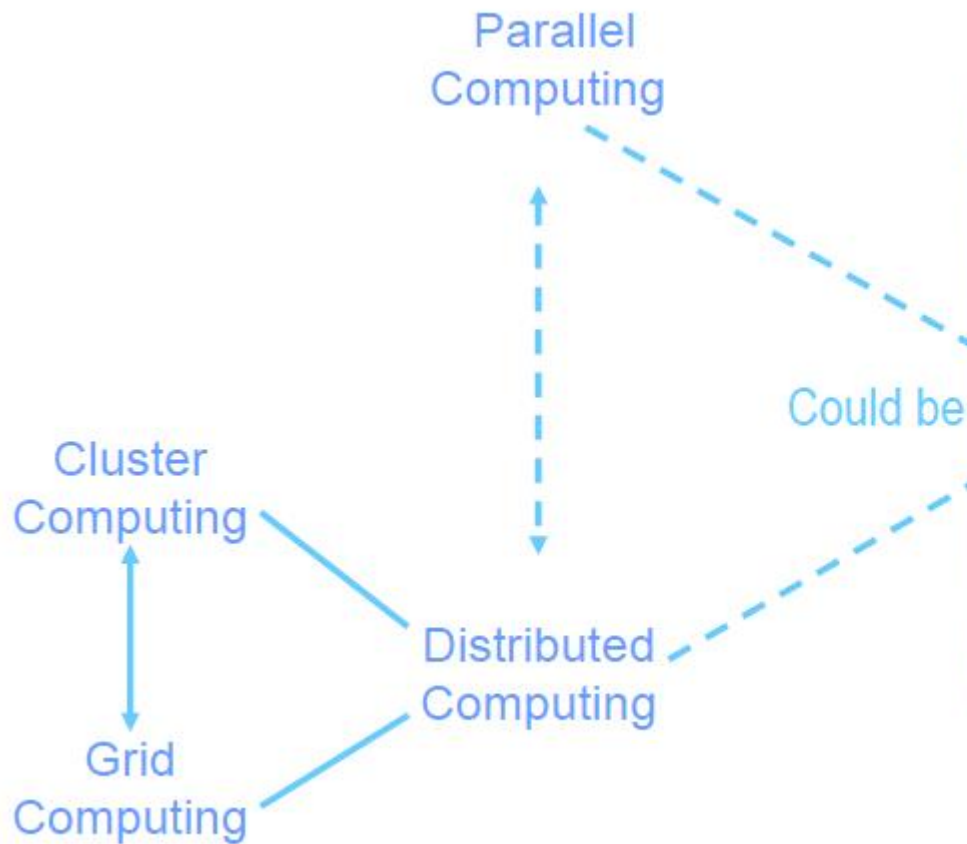
▶ Mobile Computing

Green Computing Class

- **Ubiquitous**= “seeming to be in all places”
- **Pervasive**= “present or noticeable in every part of a thing or place”
- Information processing engaged in everyday's activities and objects.
- Term used since 1980s
- Different models but same vision:
 - Small, inexpensive, robust devices distributed throughout everyday's life



Blue Computing Class



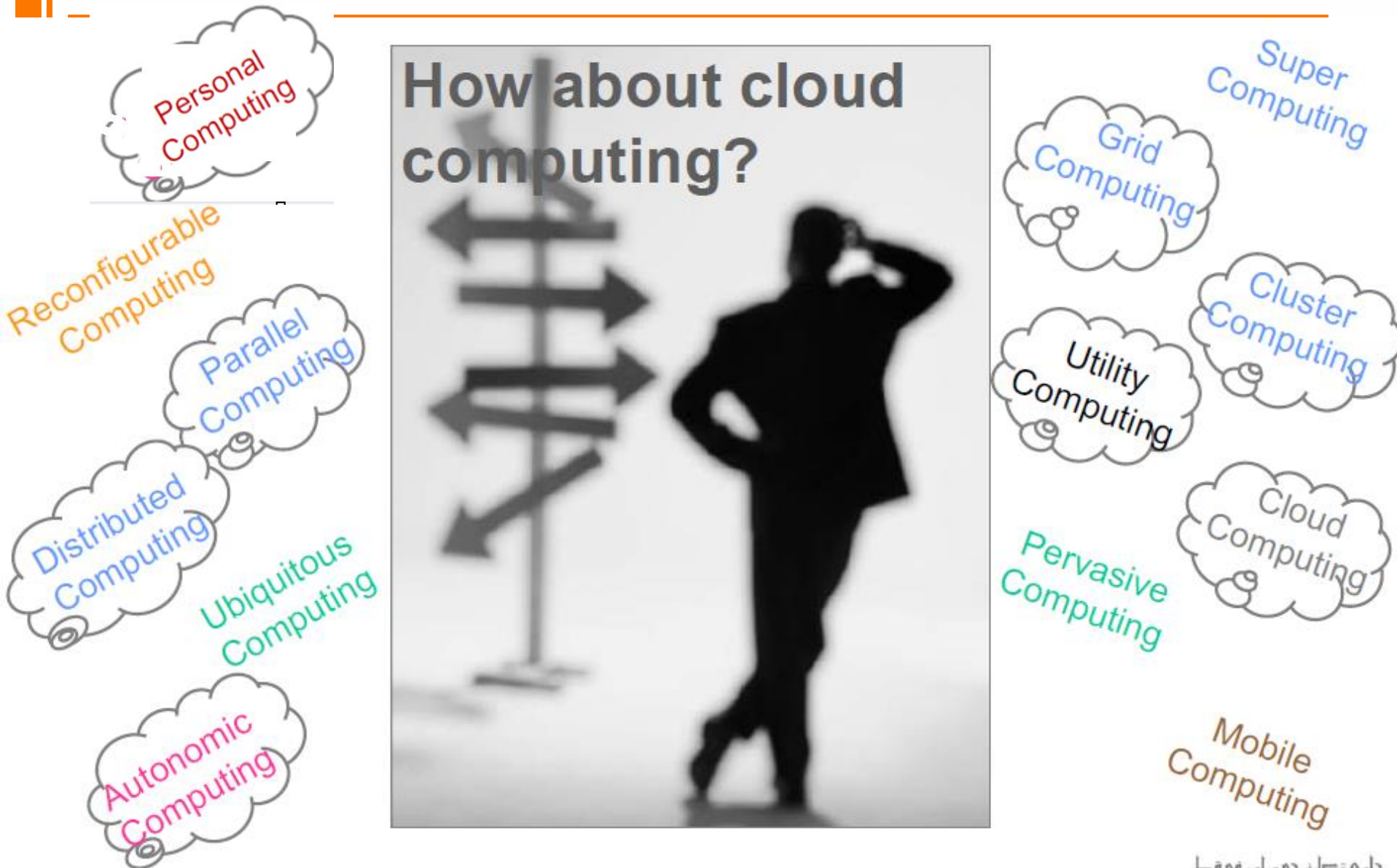
Super Computing



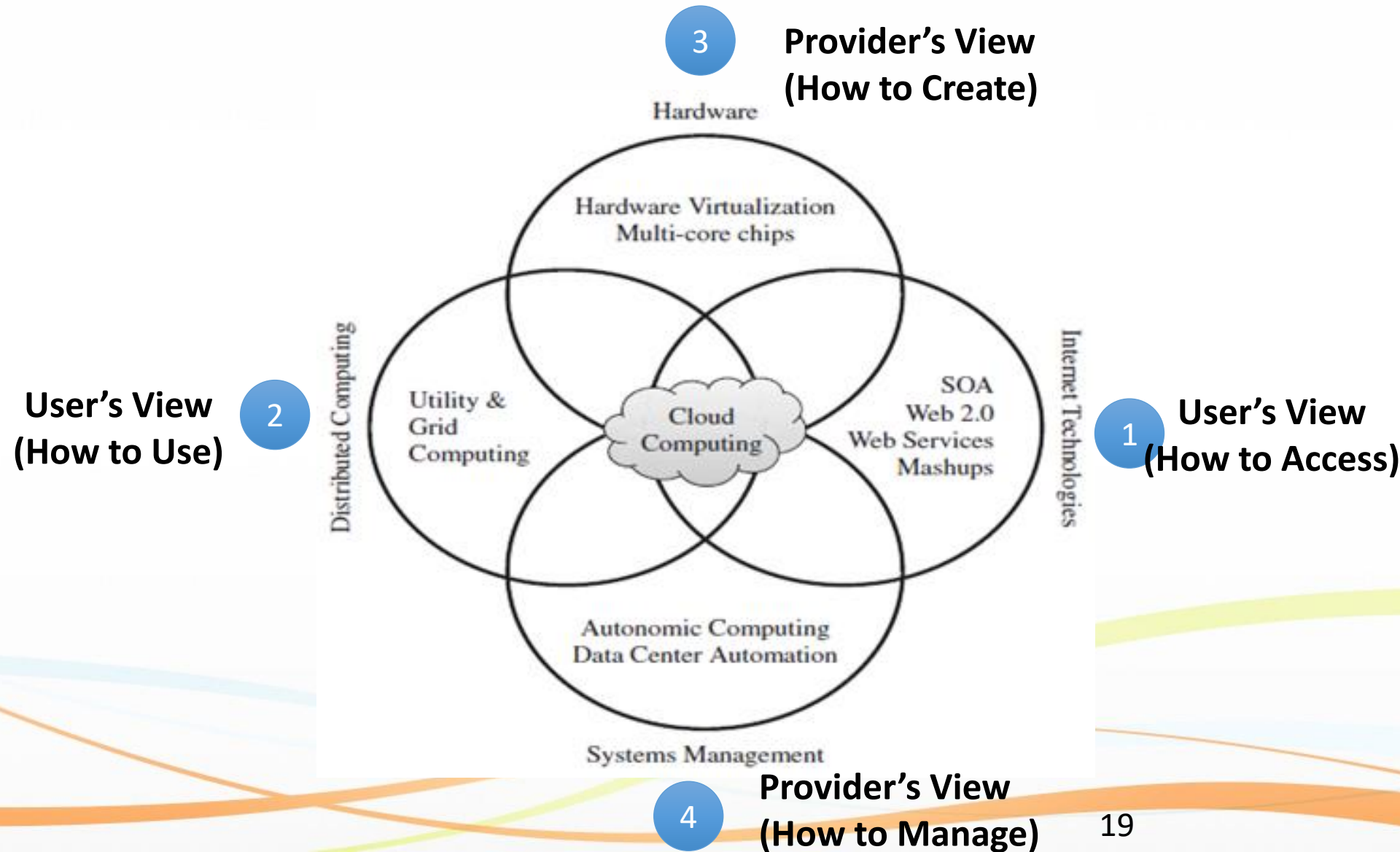
Other kinds of Computing

- ▶ Reconfigurable Computing
- ▶ Automatic Computing (Self-management)
- ▶ Utility (Service) Computing

Cloud Computing is . . .



Roots of Cloud Computing



1st Root of Cloud Computing

▶ SOA, Web Services, Web 2.0, and Mashups

❖ Web Service

- Glue together applications running on different platforms
- Information sharing

❖ SOA

- Service-oriented architecture

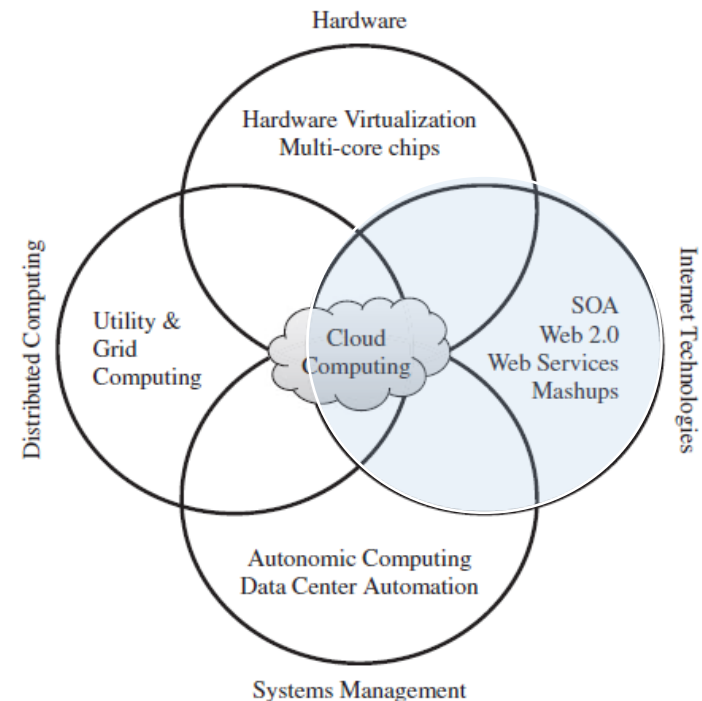


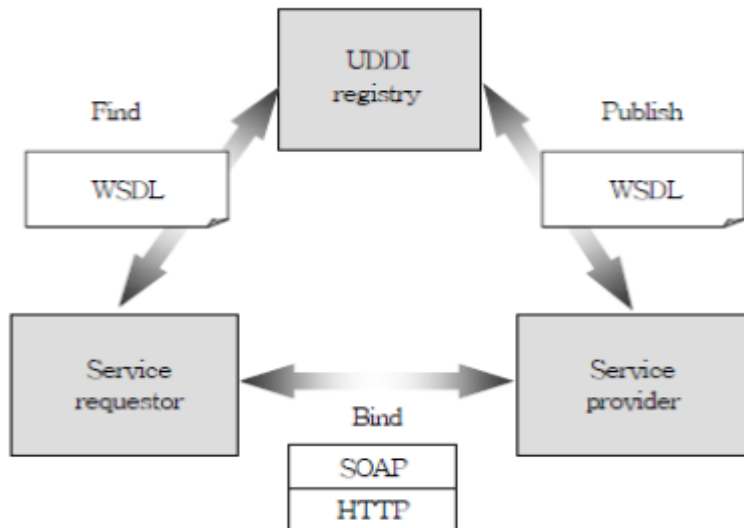
FIGURE 1.1. Convergence of various advances leading to the advent of cloud computing.

Service-Oriented Architecture (SOA)

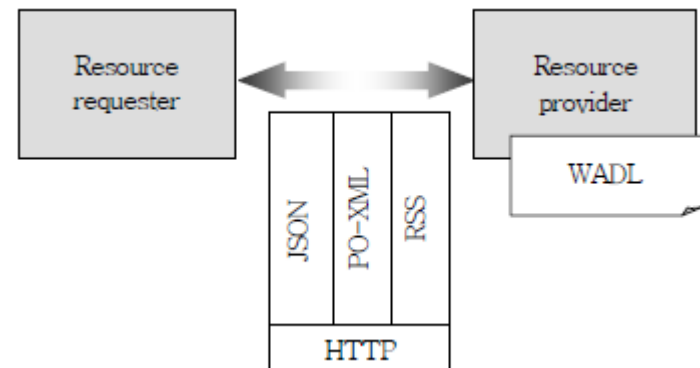
- ▶ A Service-Oriented Architecture (SOA) starts with a businesses process.
 - ❖ a service is a business functionality that is **well-defined, self-contained, independent** from other services, and published
 - ❖ used via a standard programming interface.
- ▶ Loose coupling of service orientation provides great **flexibility for enterprises** to make use of all available service resources regardless of platform and technology restrictions.

SOA Implementations in Web Services

- XML Web Services



- RESTful Web Services



- Based on XML/HTTP
- Hard to learn and use
(SOAP, WSDL, UDDI, WS-*)
- Machine friendly (M2M)

- Based on basic Web tech
(HTTP)
- Easy to learn and use
- Human friendly

SOA: Advantages

❖ Loosely-coupled connections :

- Loose-coupling is the key attribute of service-oriented architecture.
- Each service component is independent due to the stateless service feature.
- The implementation of a service will not affect its application as long as the exposed interface is not changed. This makes SOA software much easier to evolve and update.

❖ Interoperability :

- Technically, any client or service can access other services regardless of their platform, technology, vendors, or language implementations.

❖ Reusability :

- Any service can be reused by any other service. Because clients of a service need only to know its public interfaces, service composition and integration become much easier. This makes SOA-based business application development much more efficient in terms of time and cost

❖ Scalability :

- Loosely-coupled services are easy to scale. The coarse-grained, document-oriented, and asynchronous service features enhance the scalability attribute.

2nd Root of Cloud Computing

▶ Grid Computing

- ❖ Aggregation of distributed resources and transparently access to them

▶ Utility Computing

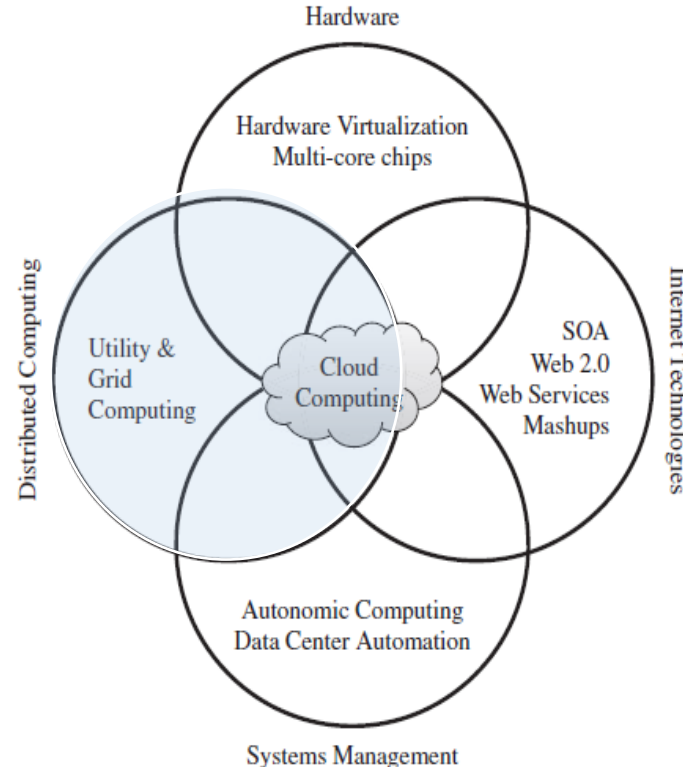
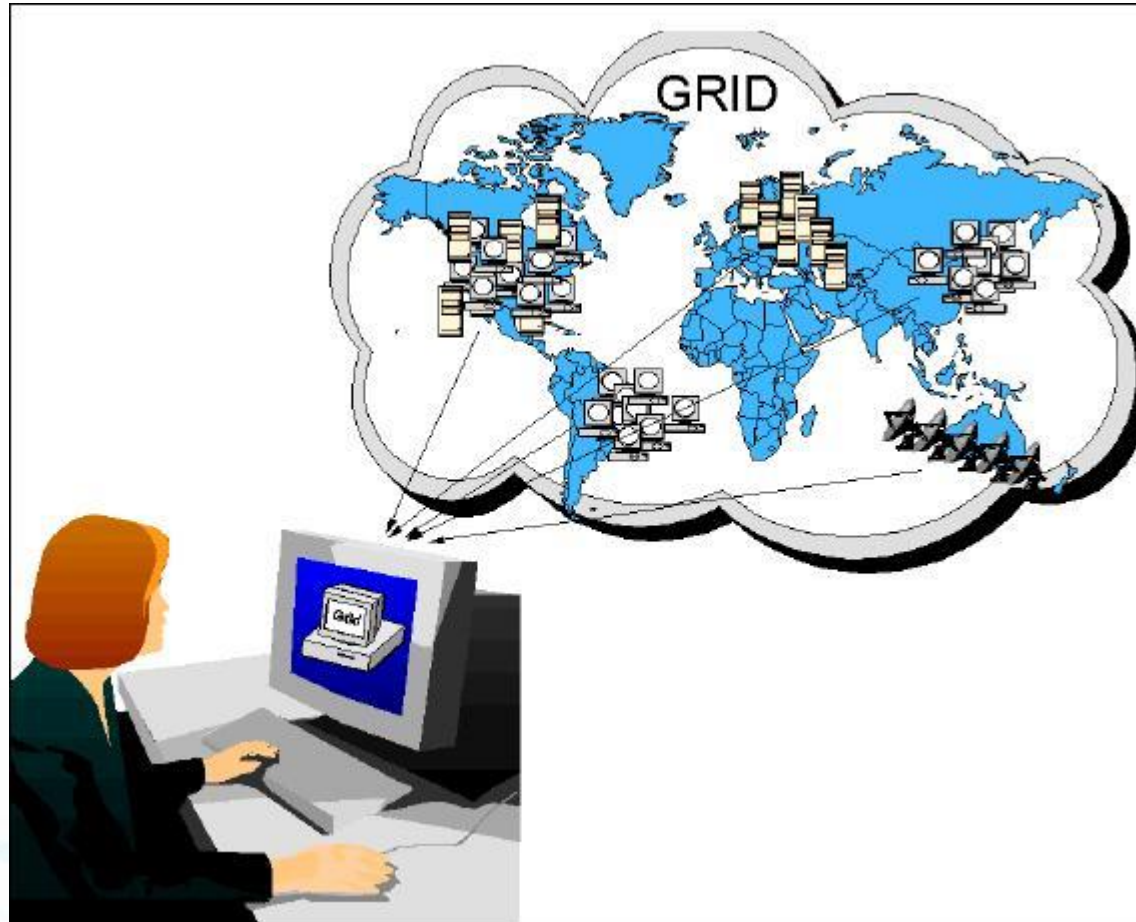
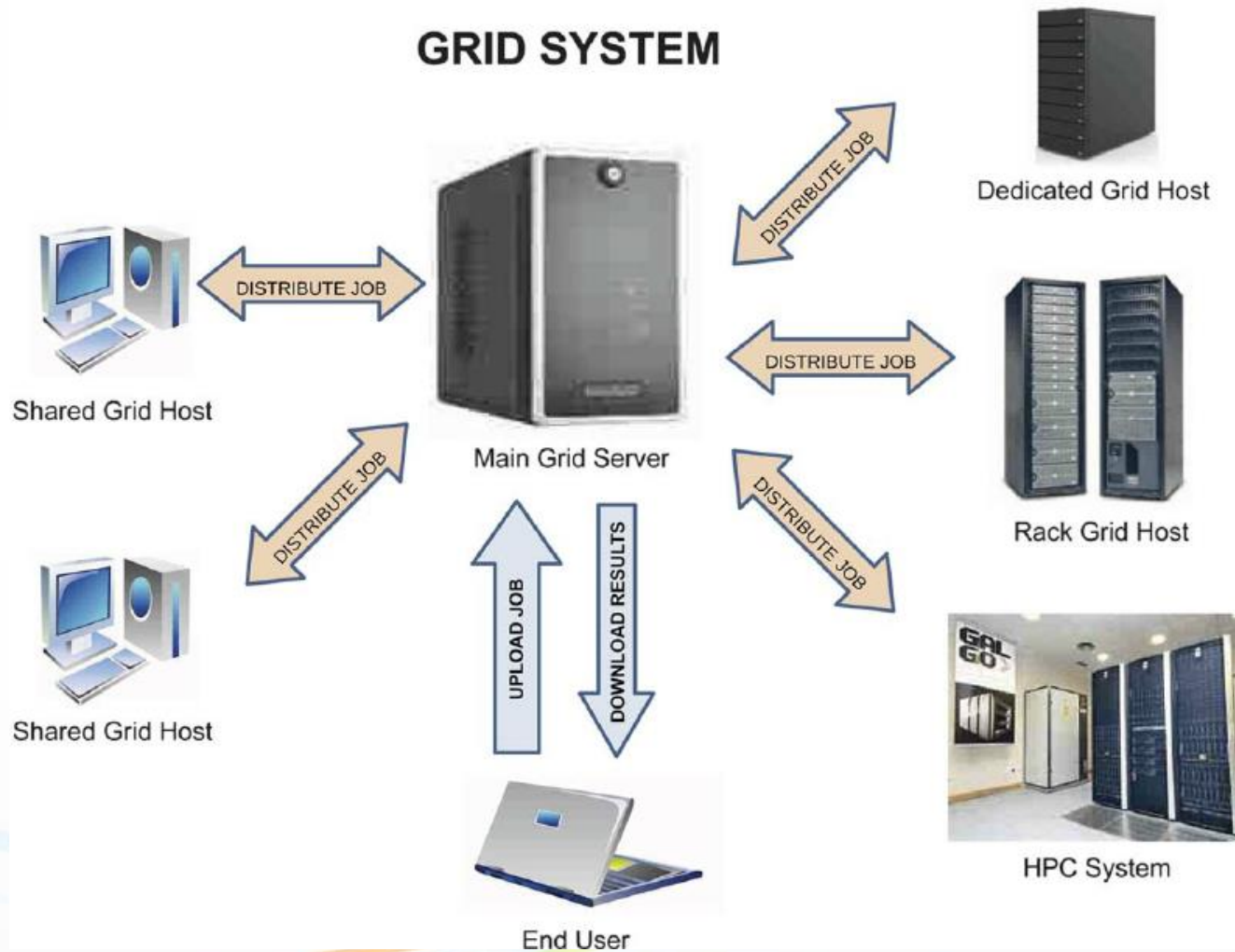


FIGURE 1.1. Convergence of various advances leading to the advent of cloud computing.

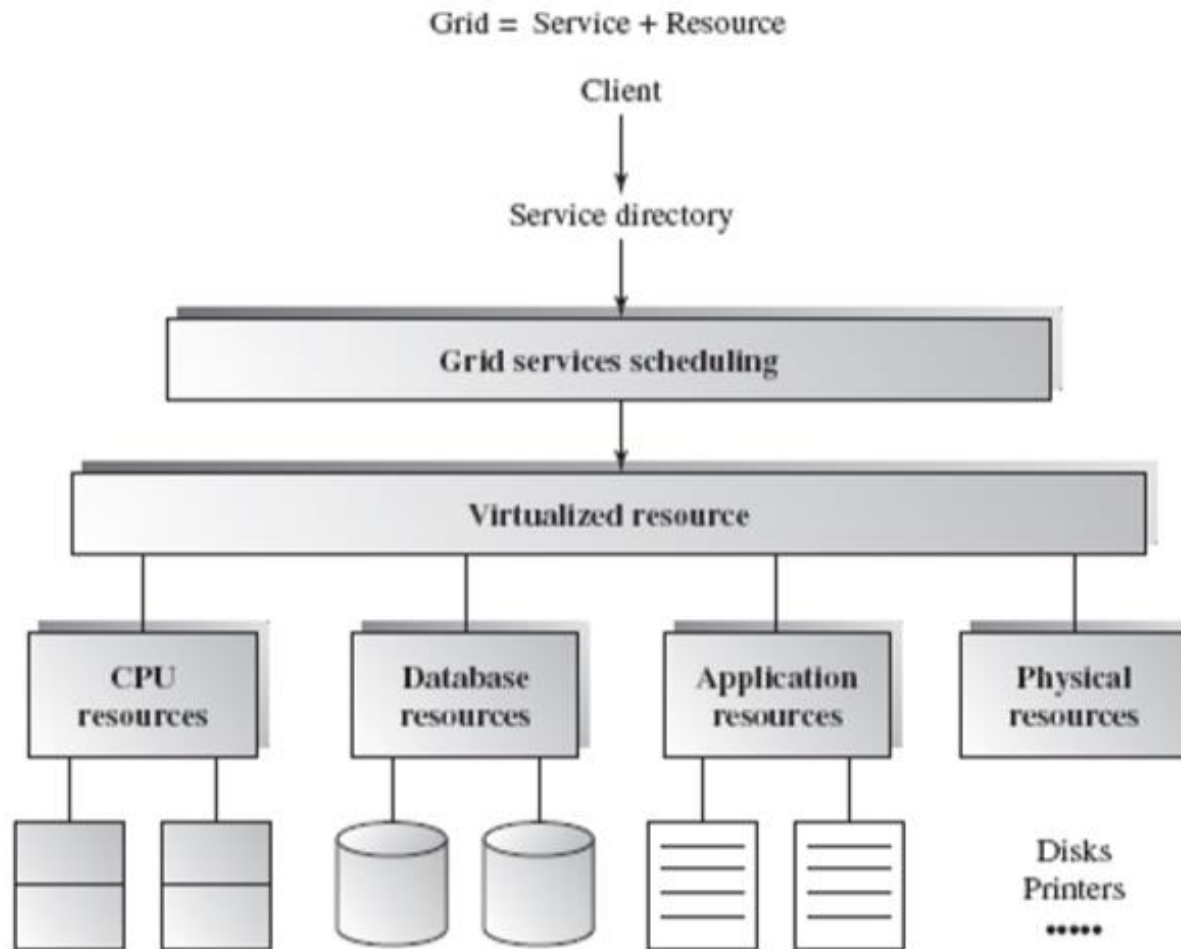
Grid Computing



Grid Computing



Grid Service Architecture



3rd Root of Cloud Computing

► Hardware Virtualization

- ❖ Hardware virtualization can be considered as a perfect fit to overcome most operational issues of data center building and maintenance

► Virtual Appliances and the Open Virtualization Format

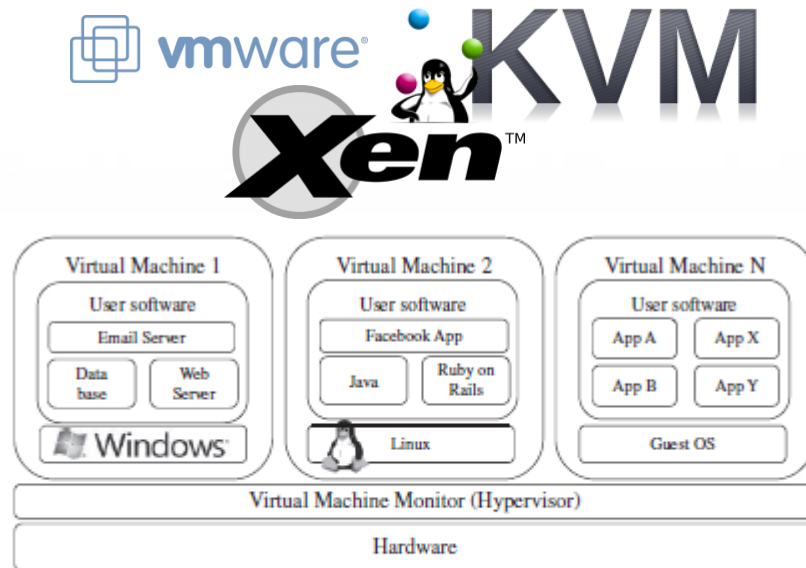


FIGURE 1.2. A hardware virtualized server hosting three virtual machines, each one running distinct operating system and user level software stack.

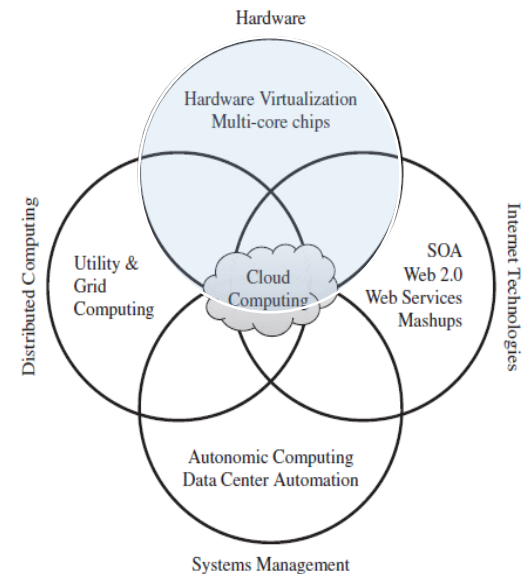


FIGURE 1.1. Convergence of various advances leading to the advent of cloud computing.

Implementation Level of Virtualization

▶ HAL Level Virtualization

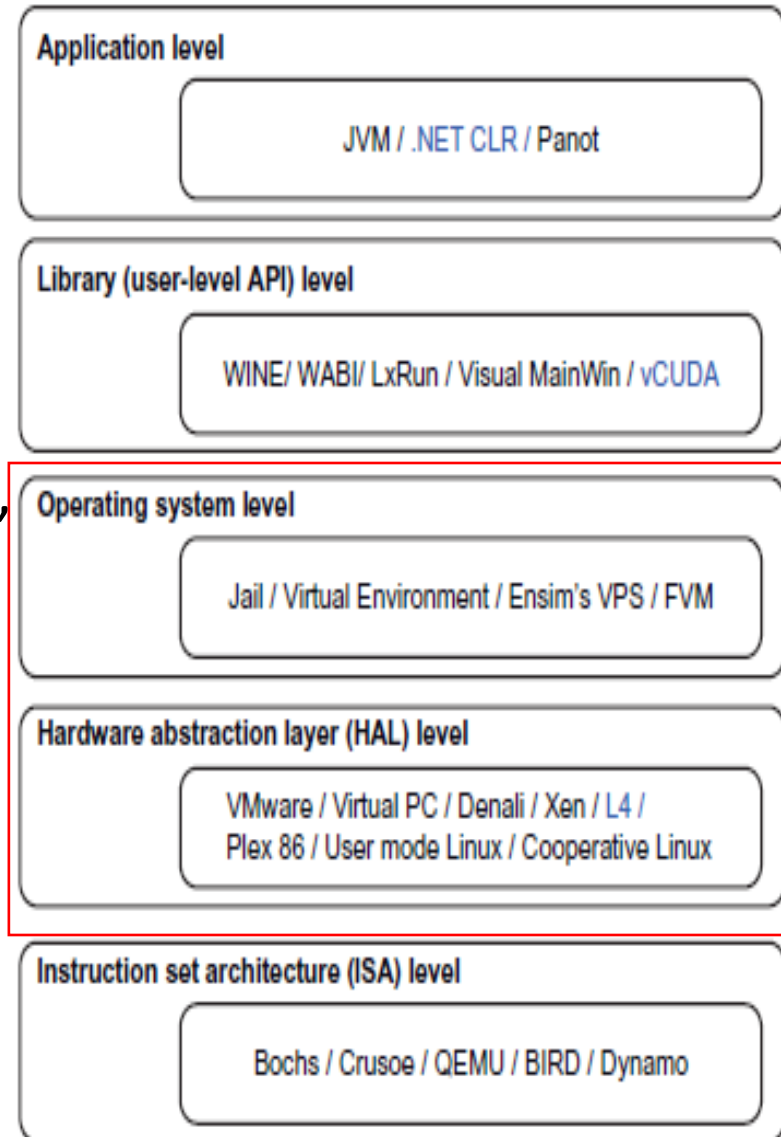
❖ **Hardware Abstraction Layer**
virtualization is performed right
on top of the bare hardware

❖ **Virtualizing a computer's
resources**, such as its processors,
memory, and I/O devices

❖ Examples

- Xen
- KVM(Kernel-Based VM)
- Vmware

=> **Hardware Virtualization**



4th Root of Cloud Computing

▶ Autonomic Computing

❖ Self-configuration

- Automatic configuration of components

❖ Self-optimization

- Automatic monitoring and control of resources to ensure the optimal functioning with respect to the defined requirements

❖ Self-healing

- Automatic discovery, and correction of faults

❖ Self-protection

- Proactive identification and protection from arbitrary attacks

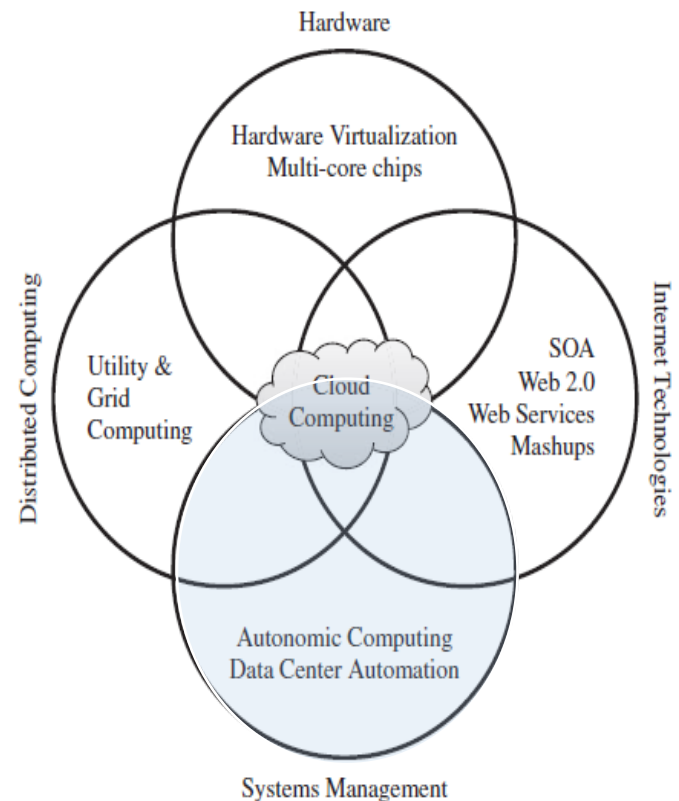


FIGURE 1.1. Convergence of various advances leading to the advent of cloud computing.

What is Autonomic Computing?

▶ Autonomic

- ❖ Pertaining to operating system that responds **automatically** to problems or system failures.

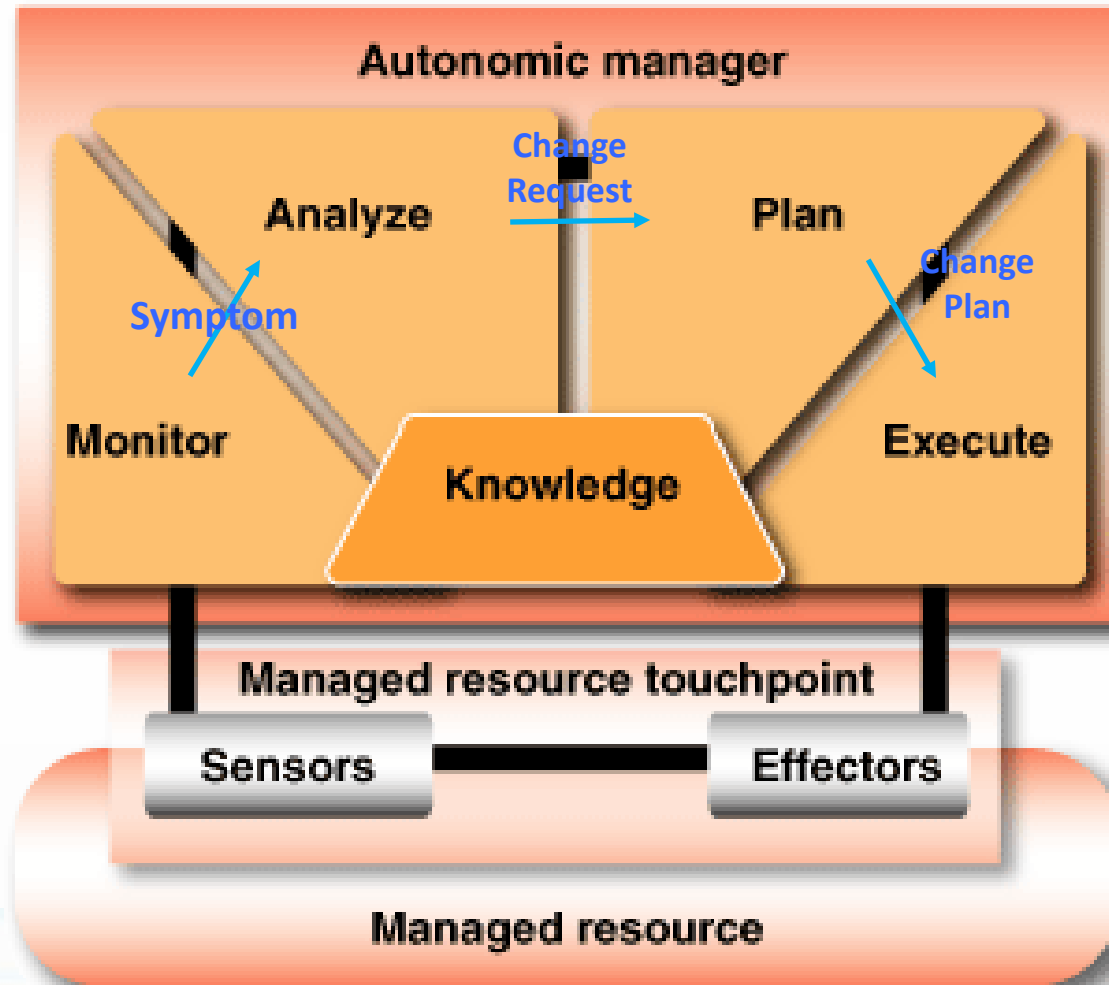
▶ Autonomic Nervous System

- ❖ handles many crucial (but what we'd consider mundane) functions without requiring any conscious thought on our part.
- ❖ When we run, it increases our heart and breathing rates. If we get too hot, it redirects blood flow to "cool us". If we turn a light on or off, it adjusts our pupils for maximum visual activity.
- ❖ This leads us to describe an approach we call "autonomic computing."

▶ Autonomic computing

- ❖ A **computing environment** with the ability to manage itself and dynamically adapt to change in accordance with business policies and objectives

Autonomic Computing



Autonomic Computing Attributes

Increased Responsiveness

Adapt to dynamically changing environments

Business Resiliency

Discover, diagnose, and act to prevent disruptions

Operational Efficiency

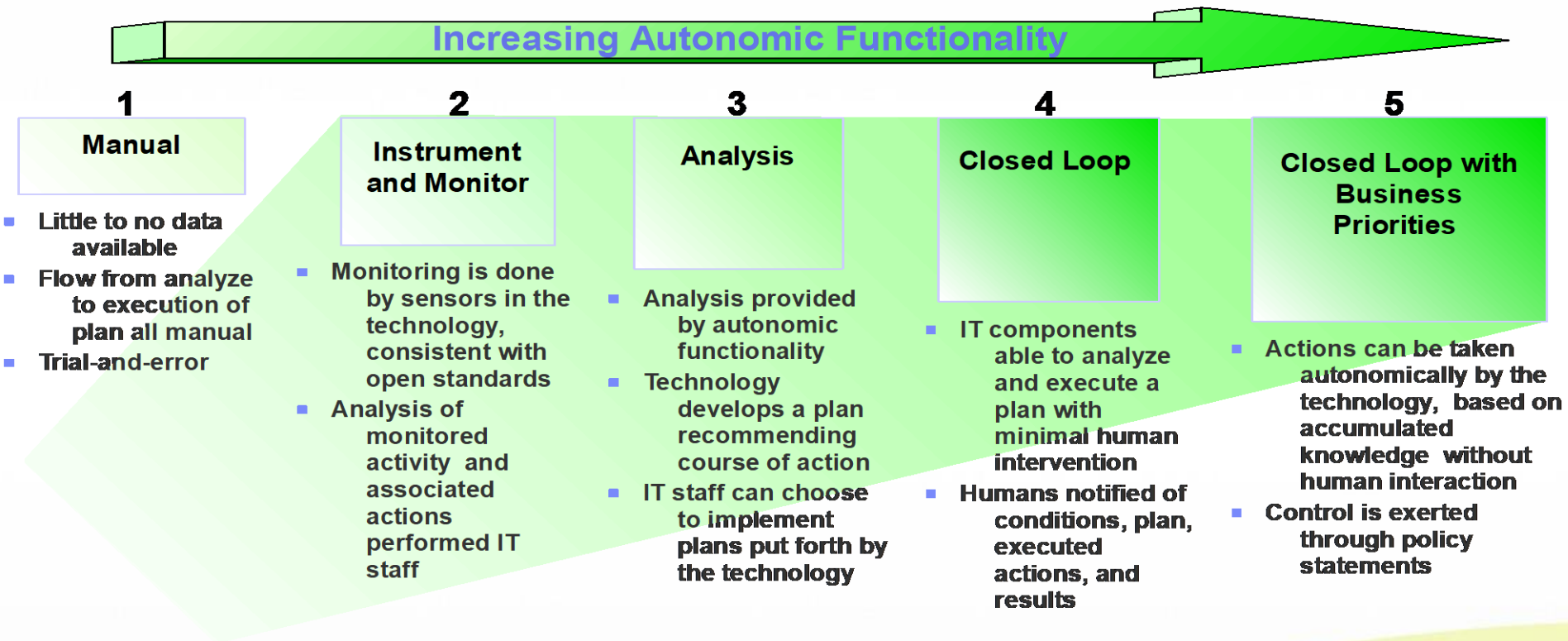
Tune resources and balance workloads to maximize use of IT resources



Secure Information and Resources

Anticipate, detect, identify, and protect against attacks

Evolving towards Autonomic Computing Systems



Definition of Cloud Computing

- ▶ “Cloud is
 - ❖ a **parallel and distributed computing system**
 - ❖ consisting of a **collection of inter-connected and virtualized computers**
 - ❖ that are **dynamically** provisioned and
 - ❖ presented as **one or more unified computing resources**
 - ❖ based on service-level agreements (SLA) established through negotiation between the service provider and consumers.”

(Prof. Buyya in Cloud Computing)

Definition of Cloud Computing

▶ “a **pay-per-use model**

- ❖ for enabling available, convenient, on-demand network access
- ❖ to a **shared pool of configurable computing resources** (e.g. networks, servers, storage, applications, services)
- ❖ that can **be rapidly provisioned and**
- ❖ **released with minimal management effort or service provider interaction.”**

(By The National Institute of Standards and Technology (NIST))

Characteristics of CC to Customer

► Pay-per-use(Metered Service)

- ❖ Cloud systems **automatically control and optimize resource use by leveraging a metering capability** at some level of abstraction appropriate to the type of service
- ❖ Resource usage can be **monitored, controlled, and reported, providing transparency for both the provider and consumer** of the utilized service.

Characteristics of CC to Customer

▶ Elasticity and the illusion of infinite resources

- ❖ Capabilities can be **elastically provisioned and released**, in some cases automatically, to scale rapidly outward and inward commensurate with demand
- ❖ To the consumer, the **capabilities** available for provisioning often **appear to be unlimited** and can be **appropriated in any quantity at any time**

Characteristics of CC to Customer

► Self-service interface

- ❖ A consumer can **unilaterally provision computing capabilities**, such as server time and network storage, as needed **automatically without requiring human interaction** with each service provider

► Resources that are abstracted or virtualized

- ❖ The provider's **computing resources** are **pooled to serve multiple consumers** using **a multi-tenant model**, **with different physical and virtual resources** dynamically assigned and reassigned according to consumer demand

Characteristics of CC to Customer

► Broad network access

❖ **Capabilities** are **available over the network** and **accessed through standard mechanisms** that promote use by heterogeneous thin or thick client platforms (*e.g., mobile phones, tablets, laptops, and workstations*)

Summary

▶ 4 Root Techs of Cloud Computing

Web Service, Grid/Utility Computing, Virtualization, Autonomic Computing

▶ Definition of Cloud Computing

Dynamic Provisioned Virtualized Distributed Systems

▶ Characteristics of Cloud Computing

Pay Per Use, Elastic Provisioning, Unlimited Resource, Virtualized Resource, Broad Access



Thank You!!

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Any Questions ??